

Relationship	From	To	Cardinality
Sends	Users	Transactions	0..1 - 0..M
Listed in	Transactions	Transaction_Category	1 - 0..M
Applies to	Transaction_Categories	Transaction_Category_Map	1 - 0..M
Generates	Transactions	Sysrtem_Logs	1 - 0..m
Receives	Users	Transactions	

Users		Data Type
PK	user_id	INT, AUTO_INCREMENT
	phone_numbers	VARCHAR(15), NOT NULL, UNIQUE
	first_name	VARCHAR(50), NOT NULL
	last_name	VARCHAR(50), NOT NULL
	national_id	VARCHAR(20), UNIQUE, NULL
	account_type	ENUM('personal','business','agent')
	registration_date	DATETIME, NOT NULL

Transaction_Categories		Data Type
PK	category_id	INT, AUTO_INCREMENT
	category_name	VARCHAR(50), NOT NULL, UNIQUE
	category_type	VARCHAR(30), NOT NULL
	description	VARCHAR(255), NULL

Transactions		Data Type
PK FK FK	transaction_id	INT, AUTO_INCREMENT
	transaction_reference	VARCHAR(50), NOT NULL, UNIQUE
	sender_id	INT, NULL → Users.user_id
	receiver_id	INT, NOT NULL → Users.user_id
	amount	DECIMAL(12,2), NOT NULL, >0
	currency	VARCHAR(3), NOT NULL, DEFAULT 'RWF'
	transaction_date	DATETIME, NOT NULL
	balance_after	DECIMAL(12,2), NULL
	status	ENUM('pending','completed','failed','reversed')
	tests	

Transaction_Category_Map (junction)		Data Type
PK FK FK	map_id	INT, AUTO_INCREMENT
	transaction_id	INT, NOT NULL → Transactions.transaction_id
	category_id	INT, NOT NULL → Transaction_Categories.category_id

Systems_Logs		Data Type
PK FK	log_id	INT, AUTO_INCREMENT
	transaction_id	INT, NULL → Transactions.transaction_id
	log_timestamp	DATETIME, NOT NULL
	process_stage	VARCHAR(50), NOT NULL
	status	ENUM('success','warning','error')
	message	VARCHAR(255), NULL

